REP-107 · Release A, first two packages: flow repairs built, and the balance after DEC-101

Game	Fruit Market 1.19.1 live; rules version 18 on the working branch (not released)
Report date	2026-11-12 (Asia/Kuala_Lumpur)
Type	Build and results report
Request	DEC-098 (three releases; release A in packages) and DEC-101 (Charmer 3 favour cards; Ripening Room without its discount; mango orders left)
Approval	PLAN-081 and PLAN-082 approved and built. Publishing: not requested, not done.

1. PLAN-081: flow repairs and two defects

Built by a helper in a separate work folder, then checked and merged by me (merge `f50c9d5`). Every finding from the flow study was confirmed in a running game before it was fixed; none was wrong.

Item	Before	Now
Reduced motion	The open crates showed for 0.15 s and "No sales today" for 0.06 s	2.9 s and 1.1 s, as with motion on; a tap still ends the pause
"Skip" in the rush	A tap on Skip only moved to the next customer	Skip reaches the day's takings
Hurry	Three rivals took 13 s to buy and a tap changed nothing	12 s untouched, under 4 s after one tap on the shelf or the lanes
Market Day bid	The customer row emptied; the panel said "Bid in" before you bid	The row stays; the panel says the other stalls bid first until it is your turn
Night	With "Fast rivals" the panel closed before the slide ended	One clock: the slide always finishes
Lorry	A 0.7 s wait for a lorry that is not drawn	The fruit leaves at once
Solo browser test	It could get stuck at the Favoured Trade step without failing	It plays the step and fails if the results are not reached
A paused game	The Daily market, the Market League and New game replaced it without asking	A question first: "Keep my game" or "Start anyway"
Trophy names	Names with a space were dropped at every account sync	Kept. Progress already lost cannot be restored

Two things to know:

- A tap's hurry does not carry into the rush. Otherwise a tap made while rivals set prices would skip the bell and every customer card.
- With the customer row kept, the bid panel needs a scroll to reach the price and Pass on phones 640 to 699 px tall. That panel is redrawn in package 3.

2. PLAN-082: the two rule changes

- Ah Chen (the Charmer) starts with 3 favour cards. She still holds up to 4 and earns up to 2 a day.
- The Ripening Room costs $4 and ripens green mangoes in one night. It no longer makes them $1 cheaper.
- Computer players now value the room by a fresh measurement. With no rival holding one it adds $1.54 a game for a Normal computer (800 games) and $2.81 for a Hard one (600 games). It was about $7.70 with the discount.
- The rules version stays 18: it has not shipped and is not on the preview.
- One existing test was changed on purpose. The check that the pricing hint "Enough buyers" agrees with the bell did not allow for pieces a stall delivers to its own standing order. The new numbers changed which seeded games it plays and exposed that. The hint was right; the test was corrected.

3. Balance with both changes

All-computer games, 10 days, played with the scripts in `docs/studio/review-17/balance/scripts18/` on commit `9c0d11b`. Points away from a fair share of wins:

Shopkeeper	**2 stalls**	**3 stalls**	**4 stalls**	**6 stalls**	**3 stalls, Hard**	**4 stalls, Hard**
Rosa, Haggler	-3.2	-3.6	-0.5	-0.3	-2.2	-3.5
Uncle Benny, Big Basket	-2.0	-2.8	-0.7	-3.0	-3.2	+0.2
Raj, Bargain Hunter	+2.8	+1.7	-0.7	+2.4	**+5.6**	**+4.8**
Ah Chen, Charmer	+2.3	+3.7	+1.8	+1.2	+2.2	-0.6
Kiki, Tip Jar	+3.2	+0.3	-1.9	-2.0	+1.4	+0.2
Saleh, Nest Egg	+0.2	-0.6	+3.9	**+4.3**	-0.2	+0.4
Pak Din, Shade Cloth	-1.7	-0.3	-1.5	-3.8	-1.7	-1.8
Farah, Sharp Ears	-1.4	+1.6	-0.5	+1.3	-1.8	0.0
Largest distance	3.2	3.7	3.9	4.3	5.6	4.8
Before these changes	4.5	3.3	4.0	4.6	3.9	4.2
Games	3,600	2,800	2,000	2,600	1,600	1,200
Each figure is known to about	1.7	1.5	1.4	0.9	2.0	1.8

What it shows:

- **The Charmer is fixed.** She was above fair by 4.5, 2.6, 4.0 and 4.6; she is now at +2.3, +3.7, +1.8 and +1.2.
- **Normal computers: within 4 points at 2, 3 and 4 stalls.** At 6 stalls Saleh's Nest Egg is +4.3, just past the line. I played a second 6-stall batch to check; the figure is the two together.
- **Hard computers: Raj's Bargain Hunter is +5.6 and +4.8.** He is above fair at five of the six tables. With the room's discount gone, his $1 off the first fruit each day counts for more.
- 45 of 48 results are within 4 points. The three that are not are each close enough to the line that chance alone could explain them, but Raj's pattern across tables looks real.

The Ripening Room is now a choice.

	2 stalls	3 stalls	4 stalls	6 stalls	3 Hard	4 Hard
Stalls that built it, before	90%	75%	64%	50%	71%	60%
Stalls that built it, now	16%	18%	18%	17%	38%	34%

What each upgrade adds to a stall's final cash against never building, 3 stalls, rivals choosing freely:

Upgrade	Normal (500 games a row)	Hard (400 games a row)
Juice Bar $6	+$3.69	+$3.19
Ripening Room $4	+$2.42	+$1.26
Order Book $3	+$1.73	+$0.97
Premium Display $5	+$1.55	+$2.38
Handcart $6	-$1.06	-$0.11
The computer chooses	+$3.37	+$3.43

The Juice Bar is now the strongest and the Ripening Room is in the middle. The Handcart still loses a Normal computer money, as before.

4. One decision for the owner

The target "every shopkeeper within 4 points" is met at 2, 3 and 4 stalls and narrowly missed at 6 stalls (Saleh) and against Hard computers (Raj).

Option	What it means
Accept it for release A (recommended)	Against Normal computers the game is fairer than before at 2, 4 and 6 stalls and about the same at 3. Raj and Saleh are watched in the first playtests with people, who do not play like computers
Tune further now	Saleh's Nest Egg to $14, or a smaller Bargain Hunter perk. Each needs another round of computer games, and a dollar is a blunt tool: an earlier test moved Nest Egg by about 1 point a dollar

5. Checks

Check	Result	Evidence
`npm run verify` after PLAN-081	PASS	Run by me on merge `f50c9d5` : 545 tests passed, 1 skipped
`npm run verify` after PLAN-082	PASS	Run by me: 61 test files, 545 passed, 1 skipped; lint, types and build pass
`npm run test:e2e` on PLAN-081 alone	PASS	Run by the helper: solo game with no errors; performance budget OK; online match 0 leaks; quick match; reconnect 5 of 5

Check	Result	Evidence
`npm run test:e2e` on both packages	See the PLAN-082 record	Run by me after this report was drafted
Balance target, Normal computers	PASS at 2, 3, 4 stalls; FAIL at 6 (Nest Egg +4.3)	Section 3
Balance, Hard computers	Reported	Bargain Hunter +5.6 and +4.8
Played by people	NOT RUN	
Seen by the owner on the preview	NOT RUN	Nothing has been merged to main